

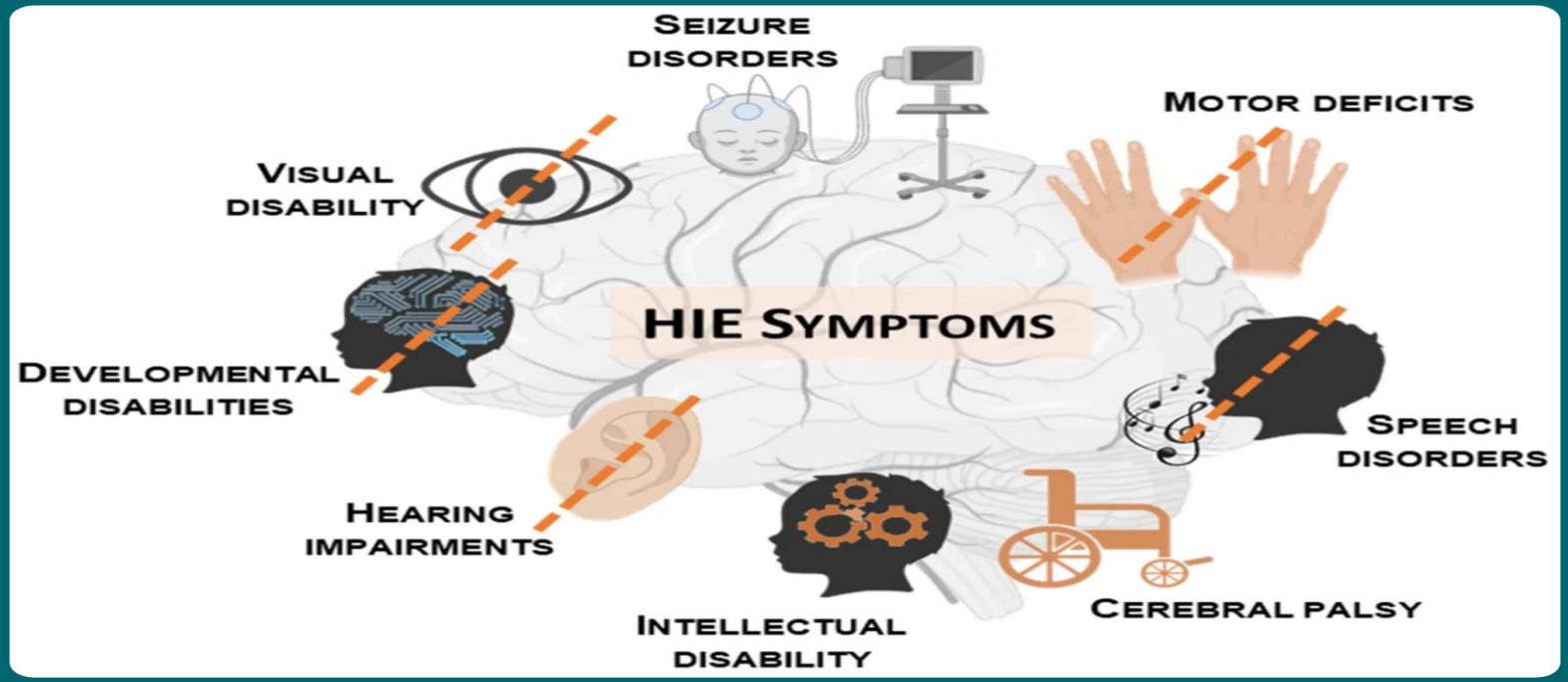
جامعة أم القرى
UMM AL-QURA UNIVERSITY

Department of Biochemistry

Faculty of Medicine – Umm Alqura University

Module : Neuroscience & Behavior

Third Year – MBBS program



Metabolic Integration of Central Nervous System

Objective

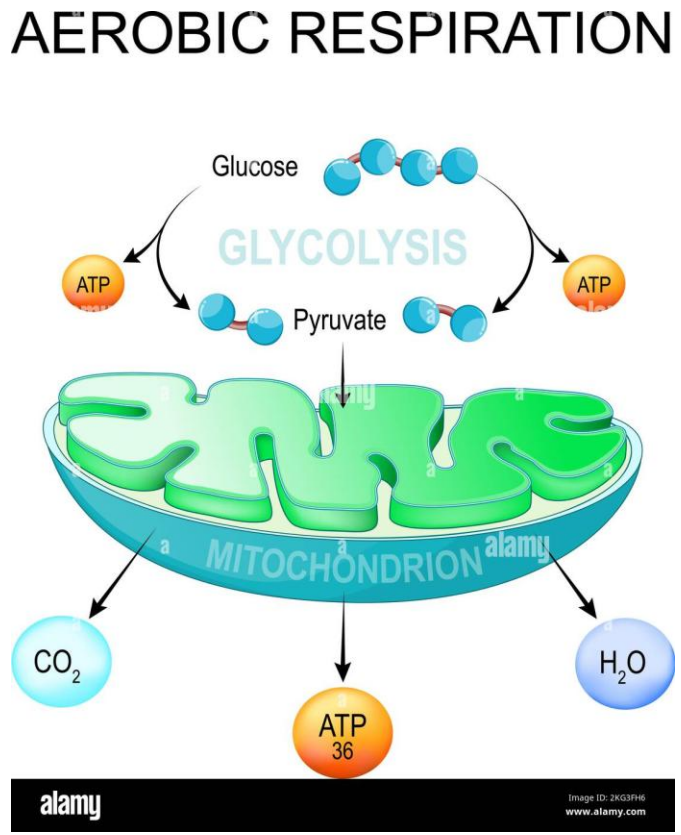
By the end of this lecture , the student will be able to:

- Discuss the pattern of fuel utilization of CNS
- Explain the mechanism of integration of metabolism of the brain during fasting and starvation.
- Discuss clinical correlation related to disturbance of energy and metabolites : Hypoxia and ischemia ; hypoglycemia ; hepatic encephalopathy

Energy source of the brain

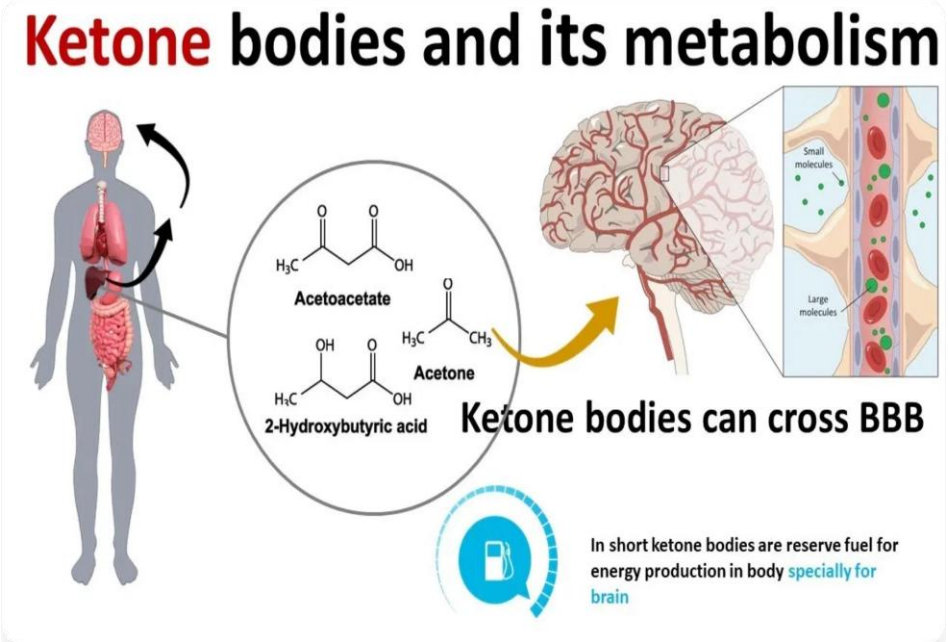
- Brain = **2% of body weight**, but uses seven times **more energy than other organs**.
- Needs **constant glucose and oxygen** supply.
- Main energy: **ATP from aerobic glucose metabolism**

  (glycolysis → pyruvate → mitochondria → ATP).



Fuels of the brain

- Glucose is the only source of fuel except during prolonged starvation.
- Ketone bodies (acetoacetate and B-OH butyrate) are produced in liver during starvation and replace glucose as fuel for the brain.
- Ketone bodies are transportable equivalents of fat as fatty acids can not traverse the Blood-Brain-Barrier because they are bound to albumin in blood.



CNS Metabolism – Glucose Dependency and Neurotransmitter Synthesis

- The brain has minimal glycogen stores and relies on a continuous glucose supply.
- The average brain consumes 120 g of glucose a day.
- Most ATP (about 70%) is used for:
  - Maintaining ion gradients (e.g., Na^+/K^+ ATPase)
  - Synthesizing neurotransmitters
  - Active transport processes
- **Glucose as a precursor for neurotransmitters:**
  - **Acetylcholine** – learning, memory, muscle control
  - **Dopamine** – reward, motivation, movement
  - **Serotonin** – mood, sleep, appetite
  - **Norepinephrine** – arousal, attention, stress response
- **GLUT3 transporter**:
  - Primary glucose transporter in neurons
  - High affinity (low Km), 5× more efficient than GLUT1
  - Ensures steady glucose supply to the brain

CNS Metabolism – Fasting and Ketone Utilization

- Due to low glycogen, the brain depends on blood glucose.
- If **blood glucose fall significantly below** a critical level for even a short time, severe and sometimes irreversible changes in brain function may result.

- **Adaptation during starvation**:
  - Brain begins oxidizing **β-hydroxybutyrate** (a ketone body) via **acetyl-CoA**
  - This shift occurs after liver glycogen is depleted
  - **Muscle protein sparing**: Ketone use delays the need for muscle breakdown for gluconeogenesis
  - In severe starvation, muscle proteins eventually become the brain's glucose source

Brain Metabolism During Fasting and Starvation

Early Fasting (0-24 hours)

Glucose remains primary fuel; liver glycogen provides glucose via glycogenolysis

Intermediate Fasting (1-3 days)

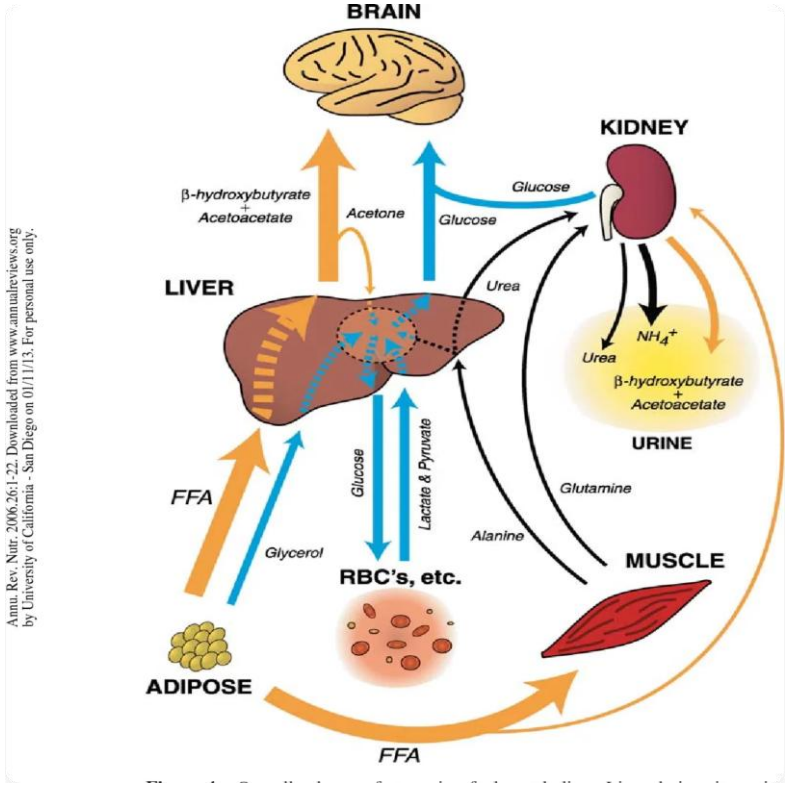
Gluconeogenesis from amino acids; initial ketogenesis begins; brain still primarily uses glucose

Prolonged Fasting (>3 days)

Ketone bodies (β-hydroxybutyrate, acetoacetate) become significant brain fuel; upregulation of ketone transporters

Extended Starvation (weeks)

Ketone bodies provide up to **60-70%** of brain energy needs; protein conservation; minimal glucose use



Annu. Rev. Nutr. 2006.26:1-22. Downloaded from www.annualreviews.org
by University of California - San Diego on 01/11/13. For personal use only.

Sources: Owen et al., 1967; Jensen et al., 2020; Puchalska & Crawford, 2017

Metabolism of CNS-Brain Lipids Synthesis & Oxidation

- **Blood-Brain Barrier (BBB)** limits entry of many fatty acids.
- Most lipids in CNS (cholesterol, sphingolipids, glycosphingolipids, cerebrosides) **must be synthesized inside the brain**.
- These lipids are essential for:
  - Neurological functions
  - Myelin synthesis by glial cells
- **Exception:** Essential fatty acids (linoleic-6 and linolenic-3) can cross BBB.
  - Inside CNS, they are elongated and desaturated to form very-long-chain fatty acids for myelin sheath synthesis.

Brain Lipids Synthesis & Oxidation

- **Peroxisomal fatty acid oxidation** is important in the brain as the brain contains very-long-chain fatty acids & branched- chain fatty acids as phytanic acid (of diet) Both are oxidized by α **oxidation** in the peroxisomes

- **Refsum disease**:
- A peroxisomal disorder that impairs metabolism of these fatty acids.
- **Leads to severe brain effects and symptoms**:
  - Peripheral neuropathy (nerve damage outside CNS)
  - Ataxia (loss of coordination)
  - Hearing loss (sensorineural deafness)



Cerebral Ischemia

- Cerebral ischemia is a medical emergency caused by an insufficient **blood supply** to the brain, leading to a lack of **oxygen** and **nutrients**, which can cause brain tissue death.

- When this insufficient blood supply results in permanent brain tissue damage **(cerebral infarction)**, the condition is known as an **ischemic stroke**, which is the most common type of stroke.

Pensato U, et al. Cerebral Infarct Growth: Pathophysiology, Pragmatic Assessment, and Clinical Implications. Stroke. 2025 Jan;56(1):219-229. PMID: 39545332.

Hypoxia → ischemia → Lactic a.; ROS…

- **In cerebral ischemia, lack of oxygen switches metabolism of glucose to the anaerobic pathway & lactic acid production**

1. **It decreases pH that may injure and inactivate mitochondria**.
2. **Lactic acid degradation of NADH** (which is needed for ATP synthesis) may also interfere with adequate post-ischemic recovery of ATP levels.
3. **Lactic acid increase the amount of free-radical mediated injury**. ↑ **Lactic acid** in neurons → acidosis → promotes the pro-oxidant effect → ↑ the rate of conversion of O_2 to H_2O_2 or to hydroxyperoxyl radical (ROS).
4. **Oxidative stress →**
   - Lipid peroxidation
   - Inactivation of enzymes
   - Nucleic acid (DNA & RNA) damage
   - Release of calcium ions from intracellular stores with more damage to neurons

The cascade of events during cerebral ischemia (lack of oxygen supply to brain neurons)

1. Oxygen supply stops → Anaerobic metabolism begins

- Neurons switch to anaerobic glucose metabolism, producing lactic acid.
- **ATP depletion** occurs because aerobic respiration stops.

2. ATP depletion → Membrane ion system fails

- Ion pumps (like Na^+/K^+ ATPase) malfunction.
- This causes **depolarization of neurons**.

3. Depolarization → Calcium influx

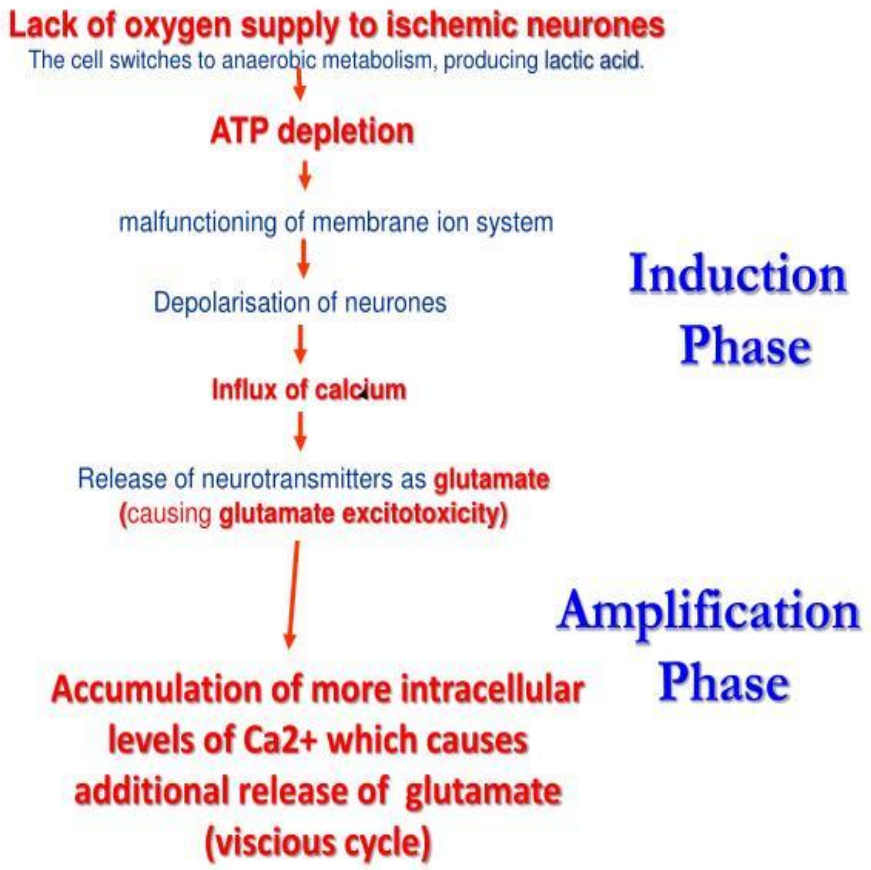
- Voltage-gated calcium channels open.
- Large amounts of **Ca^{2+} enter neurons**.

4. Calcium influx → Glutamate release

- Excitatory neurotransmitter **glutamate** is released excessively.
- This causes **glutamate excitotoxicity** (overstimulation of neurons).

5. Amplification phase

- More Ca^{2+} accumulates inside neurons.
- Additional glutamate release occurs → **vicious cycle**.
- Leads to **neuronal death**.



Glutamate Excitotoxicity

Glutamate Excitotoxicity, overstimulates neurons, leading to their death. This process plays a significant role in various neurological disorders, including stroke, Neurodegenerative Diseases like Alzheimer's disease, and Huntington's disease.

occurs when the cellular energy reserves are depleted

(as in hypoglycemia or stroke, etc)

↓

Failure of the energy-dependent reuptake pumps of glutamate

remove glutamate from the synaptic cleft failure.

↓

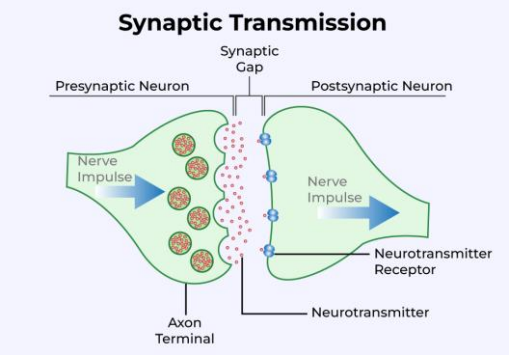
Accumulation of glutamate in the synaptic cleft

↓

Overstimulation of the postsynaptic glutamate receptors

↓

Prolonged glutamate receptor activation leads to prolonged opening of the receptor ion channel and the influx of lethal amounts of Calcium ions, which can activate cytotoxic intracellular pathways in the postsynaptic neurons



Hypoglycemic Encephalopathy

- ***As blood glucose falls below 45 mg/dL***

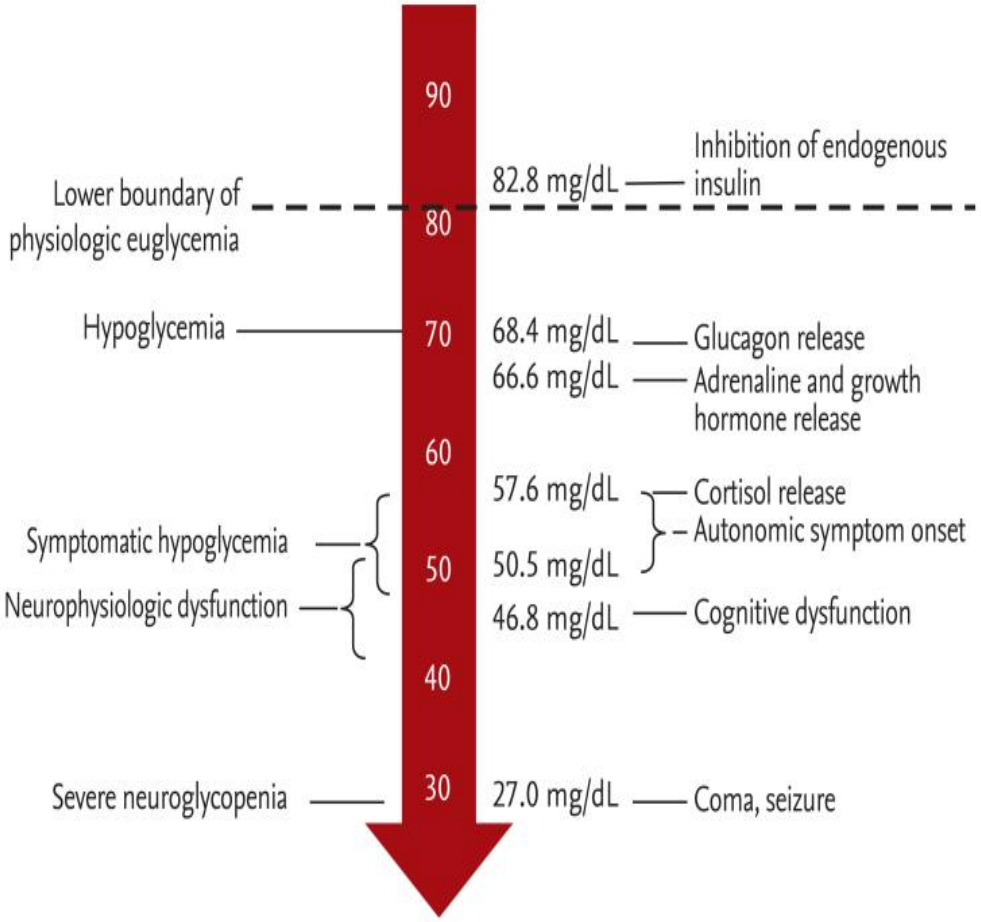
  the brain attempts to use internal substrates such as glutamate and TCA cycle intermediates as fuels for ATP production. Because the pool size of these substrates is quite small, they are **quickly depleted**.

- ***As the blood glucose drops from 45 to 36 mg/dL***

  Symptoms appear to arise from **decreased synthesis of neurotransmitters** in particular regions of the brain (hippocampal & cortical structures)

- ***If blood glucose levels continue to fall***

  Neuronal cell death ensues that may be caused by **glutamate excitotoxicity** ?? (as result of ATP depletion, nerve cells are damaged by glutamate) This occurs when receptors for glutamate receptor are over activated



Clinical Manifestation of Hypoglycemic Encephalopathy

- **Clinical manifestations of hypoglycemia:**

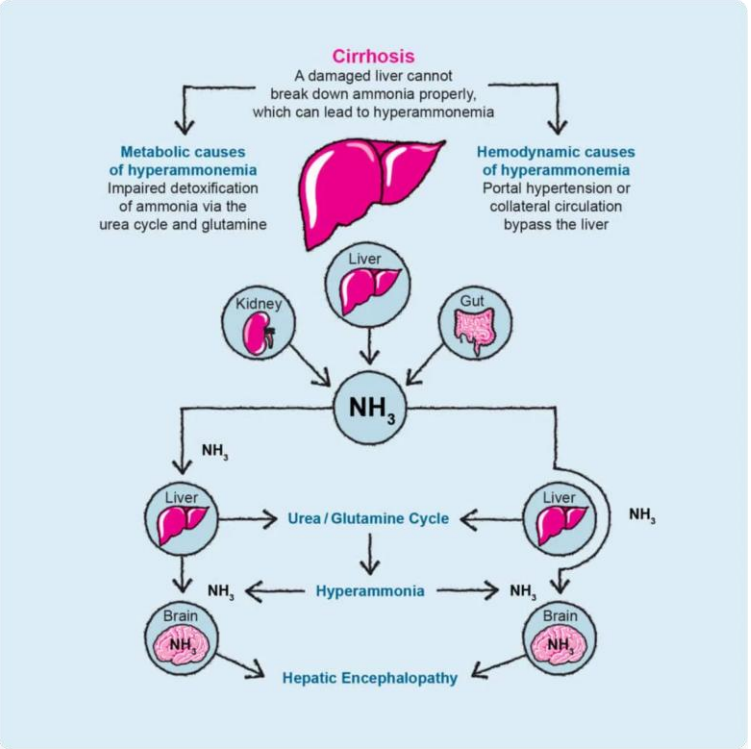
❖ Early clinical signs in hypoglycemia initiated by hypothalamic sensory nuclei as **sweating, palpitations, anxiety and hunger**.

❖ late stages, these symptoms give way to serious manifestations of **CNS disorders** as confusion, lethargy, seizures & coma

Hyperammonemia & hepatic encephalopathy

- In hepatic diseases complicated with hyperammonemia, **ammonia** can diffuse into the brain from the blood to neurons.
- The ammonia is able to **inhibit the glutaminase in neurons**, thereby decreasing formation of glutamate in presynaptic neurons.
- Glutamate is not regenerated → neurotransmission is impaired.

This effect of ammonia might contribute to the lethargy associated with the hyperammonemia found in patients with hepatic disease. (hepatic encephalopathy)



Summary – Metabolic Integration of the CNS

- **Energy Source**
  - Brain needs continuous **glucose and oxygen**.
  - Uses **aerobic oxidation of glucose** to make ATP.
  - Has **very little glycogen** reserve.
- **Starvation Adaptation**
  - During fasting, **ketone bodies** become the main fuel.
  - Helps **save muscle proteins** and maintain brain energy.
- **Lipid Metabolism**
  - **Fatty acids cannot cross** the blood-brain barrier.
  - Brain makes its own **lipids for myelin and membranes**.
  - Defect in peroxisomal oxidation → **Refsum disease**.
- **Hypoxia / Ischemia**
  - Low oxygen → **lactic acid & free radicals** damage neurons.
  - **Glutamate excitotoxicity** causes nerve cell death → stroke.
- **Metabolic Disorders**
  - **Hypoglycemia:** low glucose → confusion, seizures, coma.
  - **Hepatic encephalopathy:** high ammonia → ↓ glutamate → lethargy.

References

Textbook

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  → Chapters:
  - Chapter 58: *Integration of Metabolism*
  - Chapter 59: *Metabolism of the Nervous Tissue*
- **Devlin, T. M. (Ed.):** *Textbook of Biochemistry with Clinical Correlations*, 8th Edition.
  Wiley-Liss, 2023.
  → Sections on:
  - Brain metabolism and glucose utilization
  - Ketone body adaptation during fasting and starvation
  - Clinical correlations: hypoglycemia, ischemia, hepatic encephalopat

Thank you

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